HK IMO Training 2021-2022
Problem Set 21
For 12 March 2022

Problem 81. There is a chess piece in each cell of an $n \times n$ chessboard where $n \geqslant 2$. In each move, we simultaneously move every chess piece to a diagonally adjacent cell. Note that each cell may contain more than one chess piece. Find the minimum number of cells which contain some chess pieces after a finite number of moves.

Problem 82. Let a, b, c be nonzero real numbers satisfying $a(a^2+b+c) = b(b^2+c+a) = c(c^2+a+b) = 1$. Prove that at least two of a, b, c are equal.

Problem 83. Two circles Γ_1 and Γ_2 meet at two distinct points A and B. A line passing through A meets Γ_1 and Γ_2 again at C and D respectively. Another line passing through A meets Γ_1 and Γ_2 again at E and F respectively. Define the following intersection points.

- P is the intersection of the tangent to Γ_1 at C and the tangent to Γ_2 at D
- Q is the intersection of the tangent to Γ_1 at E and the tangent to Γ_2 at F
- S is the intersection of the tangent to Γ_1 at C and the line DF
- T is the intersection of the tangent to Γ_2 at F and the line CE
- X is the intersection of the tangent to Γ_2 at D and the line CE
- Y is the intersection of the tangent to Γ_1 at E and the line DF

Prove that the circumcircles of $\triangle BPQ$, $\triangle BST$ and $\triangle BXY$ pass through a common point different from B, or they are tangent at B.

Problem 84. Let S be a nonempty set such that every element in S is a prime number. Suppose for any distinct primes $p_1, p_2, \ldots, p_k$ in S, all prime divisors of $p_1 p_2 \cdots p_k + 1$ belong to S. Prove that S contains all prime numbers.